Exercise #1:

Consider a mortgage team with $100,000,000 to finance various investments. There are five categories of loans, each with an associated return and risk (1-10, 1 best):

Loan/investment	Return (%)	Risk
First Mortgages	9	3
Second Mortgages	12	6
Personal Loans	15	8
Commercial Loans	8	2
Government Securities	6	1

[1, 2] Quote from Quantitative Methods for the Management Science 45-760 course note by *Michael A. Trick*, Mellon University

Any uninvested money goes into a saving account with no risk and 3% return. The goal for the mortgage team is to allocate the money to the categories so as to:

(a) Maximize the average return per dollar
(b) Have an average risk of no more than 5 (all averages and fractions taken over the invested money (not over the saving account)).
(c) Invest at least 20% in commercial loans
(d) The amount in second mortgages and personal loans combined should be no higher than the amount in first mortgages.

Exercise #2:

Adapted from Winston[1] *(p.93, problem 7)*

Eli Daisy uses chemicals A and B to produce two drugs. Drug 1 must be at least 70% chemical A, and Drug 2 must be at least 60% chemical B. Up to 40 oz. of drug 1 can be sold at \$7 per oz; up to 30 oz. of drug 2 can be sold at \$5 per oz. Up to 45 oz. of chemical A can be purchased at \$6 per oz, and up to 40 oz. of chemical B can be purchased at \$4 per oz. Formulate an LP that can be used to maximize Daisy's profits.

Exercise #3:

You work for an NGO that specializes in water supply chains for efficient emergency response. Last night, an earthquake of 8 degrees Richter struck Country Omega, and your boss has instructed you to decide on the response plan to send pre-designed packages containing clean water and sanitation tools. 1 Million households are in need of assistance, of which 30% are labeled critical, 43% are labeled urgent, and 27% are labeled as optional. The name of the category relates to the time a household is expected to survive without the aid. As such, households labeled as critical must be assisted within the first 36 hours. Households labeled as either urgent or optional must be assisted within the first week for the aid to be effective. One alternative to send the packages is to ship them by air, which takes 24 hours to reach the destination and costs $0.4 per package; each plane can carry up to 50,000 packages and there is no restriction on the available number of planes. The second transportation option is by land, which takes 5 days and costs $0.2 per package; a truck has capacity to carry up to 5,000 packages, and a maximum of 80 trucks are available. The priority of the NGO is to satisfy completely the critical and urgent cases, and to deliver to at least a third of the optional households. Your maximum approved budget is $200,000; however your goal is to spend as little money as possible. Formulate an LP and determine if your problem has a feasible solution.

SOLUTIONS

Exercise #1:

Note: In this formulation, we don't use percentages (i.e., we use 9 instead of 9%), as it would not affect the optimal solution. But, in order to calculate the "realized" return, one needs to divide the objective function value of this formulation by 100.

Also, we assume that the risk in proportional to the amount invested, and therefore take the weighted average considering the money invested.

Finally, we assume that at least 20% of the total investment should be for the commercial loans, and that the total investment does not include savings.

Notation: Let the investments be numbered 1...5.

Decision Variable:

Xi = the amount invested in investment *i*.
Xs = the amount in the savings account.

Objective function:

max $9x_1 + 12x_2 + 15x_3 + 8x_4 + 6x_5 + 3x_s$

Constraints:

(1) Total amount of money available:

$$x_1 + x_2 + x_3 + x_4 + x_5 + x_s = 100,000,000.$$

(2) Average risk:

Since we want to take the average over only the invested amount, a direct translation of this constraint is

$$\frac{3x_1 + 6x_2 + 8x_3 + 2x_4 + x_5}{x_1 + x_2 + x_3 + x_4 + x_5} \leq 5$$

This constraint is not linear, but we can cross multiply, and simplify to get the equivalent linear constraint:

$$-2x_1 + x_2 + 3x_3 - 3x_4 - 4x_5 \leq 0$$

(3) Invest at least 20% in commercial loans

$$x_4 \geq 0.2(x_1 + x_2 + x_3 + x_4 + x_5)$$

or

$$-0.2x_1 - 0.2x_2 - 0.2x_3 + 0.8x_4 - 0.2x_5 \geq 0$$

(4) The amount in second mortgages and personal loans combined should be no higher than the amount in first mortgages:

$$x_2 + x_3 - x_1 \leq 0$$

(5) Nonnegativity:

Xi ≥ 0 for all i;

Xs ≥ 0

Exercise #2:

1. **Step 1:** *Define the decision variables:*

$$x_{ij} = \text{ounces of chemical } i \text{ destined for drug } j, \quad i \in \{A, B\},\ j \in \{1, 2\}$$

2. **Step 2:** *Formulate the objective function*

$$Z = \max_{(x_{ij})} \ 7(x_{A1} + x_{B1}) + 5(x_{A2} + x_{B2}) - 6(x_{A1} + x_{A2}) - 4(x_{B1} + x_{B2}) \tag{1}$$

3. **Step 3:** *Write down the constraints*

Supply:

$$x_{A1} + x_{A2} \leq 45 \qquad \text{Maximum supply of chemical A} \tag{2}$$

$$x_{B1} + x_{B2} \leq 40 \qquad \text{Maximum supply of chemical B} \tag{3}$$

Demand:

$$x_{A1} + x_{B1} \leq 40 \qquad \text{Maximum demand for Drug 1} \tag{4}$$

$$x_{A2} + x_{B2} \leq 30 \qquad \text{Maximum demand for Drug 2} \tag{5}$$

Blending requirements:

$$\frac{x_{A1}}{x_{A1} + x_{B1}} \geq 0.7 \Rightarrow x_{A1} \geq 0.7x_{A1} + 0.7x_{B1} \Rightarrow 0.3x_{A1} - 0.7x_{B1} \geq 0 \qquad \text{Minimum fraction of chemical A on drug 1}$$

$$\frac{x_{B2}}{x_{A2} + x_{B2}} \geq 0.6 \Rightarrow x_{B2} \geq 0.6x_{A2} + 0.6x_{B2} \Rightarrow 0.4x_{B2} - 0.6x_{A2} \geq 0 \qquad \text{Minimum fraction of chemical B on drug 2}$$

Nonnegativity:

$$x_{ij} \geq 0, \quad \forall i,\ j \tag{8}$$

Exercise # 3:

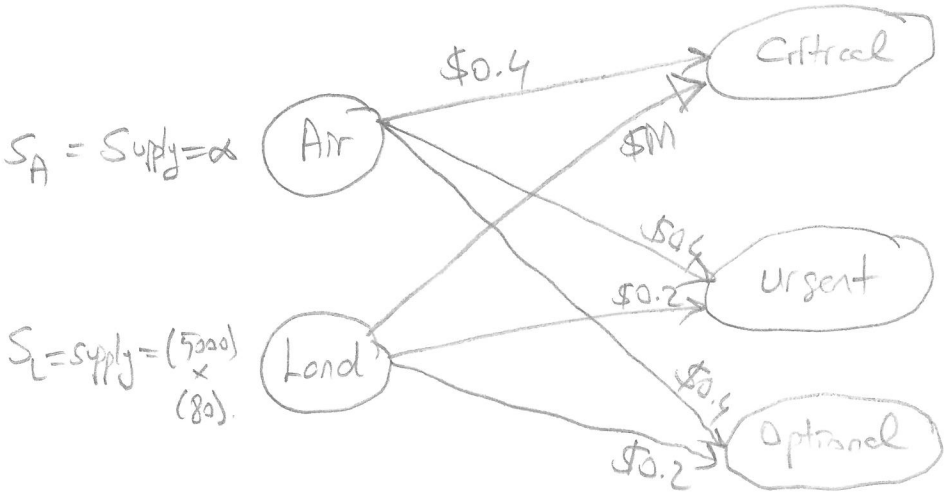
Decision Variables:

x_{ij} = # of packages to be shipped by mode i for households labeled as j

$i \in \{A, L\}$ and $j \in \{C, U, O\}$

- A → Air
- L → Land
- C → critical
- U → urgent
- O → optional.

Network Representation:



c_{ij} : cost of shipping one package from i to j.

$c_{Aj} = \$0.4$

$c_{LC} = M$ (big number)

$c_{LU} = c_{LO} = \$0.2$.

Formulation:

$$\min \quad z = \sum_i \sum_j c_{ij} \cdot x_{ij}$$

s.t.

$$\sum_i x_{ij} = d_j \text{ for } j = C, U.$$

$$\sum_i x_{ij} \geq \left(\frac{1}{3}\right) d_j \text{ for } j = O$$

$$\sum_j x_{ij} \leq S_i, \text{ for } i = A, L$$

$$\sum_i \sum_j c_{ij} \cdot x_{ij} \leq 200{,}000$$

$$x_{ij} \geq 0, \text{ for all } i, j$$

There is no feasible solution because you 200,000 is enough only to meet the demand for the critical and part of the urgent.